

MedEquiSeg: Shared Augmentation and Privileged-Prompt Reliability in Multimodal Medical Image Segmentation

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Abstract

Background: Text can provide target and spatial context for medical image segmentation, but augmentation can invalidate spatial language and target-derived prompts can exaggerate meaningful text use. Accuracy and prompt reliability therefore require separate controls.

Methods: MEDEQUISEG combines a frozen pooled BioMedCLIP adapter, two forward-active projector ATConv replacements, and a deterministic image-mask-text plan. Protocol V3 fixed manifests, split roles, mask rules, and

evaluation; the richer shared plan was a separately frozen amendment. Direction phrases are rewritten for flips and right-angle rotations; an exact no-rewrite control receives identical image-mask transforms. Five public benchmarks used three seeds, native-grid scoring, a retrospective 90-cell ablation, correct/shuffled/fixed/empty prompt controls, and a post hoc presence/class-stratified derangement.

Results: MEDEQUISEG achieved Public-5 macro Dice of **87.87%** and IoU of **80.97%**. Versus the horizontal-flip Base reference, Dice increased by **1.46** points (95% dataset-stratified bootstrap confidence interval **0.95** to **1.99**) and IoU by **1.74** points (95% interval **1.13** to **2.34**). The shared-plan no-rewrite row was numerically best at **87.96%** Dice; rewriting changed Dice by **−0.09** points (95% interval **−0.30** to **0.10**). Base plus the shared plan reached **87.83%**, leaving **0.04** points for the complete model (95% interval **−0.40** to **0.41**). Case-specific versus constant-text training differed by **1.90** points (95% interval **1.16** to **2.75**). True text exceeded presence/class-stratified derangement by **+5.63** Dice points (95% interval **+4.71** to **+6.67**), versus **+14.66** under unstratified shuffling.

Conclusions: Most macro improvement is attributable to the shared training policy; additional macro accuracy from rewriting or the BioMedCLIP-ATConv stack was not established. Presence/class matching explains part of the shuffled-prompt gap, but the residual remains privileged because public prompts contain mask- or label-derived attributes. The results support a reliability-audited benchmark, not clinical-report understanding or deployment generalization.

Keywords: medical image segmentation, multimodal learning, paired augmentation, vision-language models, privileged prompts, prompt reliability

1 Background

Vision-language pretraining has made free-text conditioning practical for dense prediction [1–3]. In medical segmentation, language can identify a target, describe morphology, or provide spatial context [4, 6, 7]. This additional modality also creates two reliability requirements that do not exist in image-only segmentation: training transforms must preserve the joint meaning of the image, mask, and prompt, and the resulting prediction should depend on the case-matched prompt when that prompt is claimed to provide useful guidance.

Conventional geometric augmentation applies one transform to the image and mask but leaves text unchanged [14]. If a prompt locates a lesion in the image frame, a horizontal flip changes *left* to *right*, a vertical flip changes *upper* to *lower*, and a right-angle rotation changes both axes. The transformed mask remains geometrically valid, yet the conditioning text may become false. This is an avoidable cross-modal supervision conflict. Separately, a geometrically valid prompt can still be uninformative or weakly grounded; high segmentation accuracy alone cannot show that the text branch is being used. The relevant augmentation example is image-frame location, rather than anatomical laterality: mirroring a breast image does not change the patient’s anatomical left or right.

We therefore treat the shared training policy as the main empirical object and prompt reliability as a control on how its accuracy is obtained. MEDEQUISEG combines biomedical text features, adaptive convolution, and synchronized image-mask-text augmentation, while matched and fixed-checkpoint controls prevent the aggregate result from being misattributed to rewriting or to clinically meaningful language understanding. Our contributions are:

1. A complete retrospective 90-cell ordered component ablation that separates the base pathway, frozen pooled BioMedCLIP embedding adapter, the effective two-projector ATConv adapter, the shared training plan, and prompt rewriting.
2. A same-data-and-evaluation-protocol comparison on five public datasets, with three training seeds, restored native-grid mask scoring, and three text-guided and six image-only architectures.
3. A protocol and prompt-reliability audit comprising an exact-plan no-rewrite control, correct/shuffled/fixed/empty inference controls, a post hoc presence/class-stratified prompt derangement, constant-text and Base-plus-plan follow-ups, prompt-provenance boundaries, and a runtime check of which registered ATConv modules enter the forward graph.

1.1 Related work

1.1.1 Text-guided segmentation

CLIP learns a transferable joint image-text space [1], and CLIPSeg adapts it to dense prediction through a lightweight conditional decoder [2]. ViLT shows that joint vision-language interaction can be built without a region detector or convolutional visual front end [5], while BioMedCLIP introduces biomedical image-text pretraining at scale [3]. LViT integrates language with a transformer-UNet pathway [6]; CausalCLIPSeg uses causal intervention for referring medical segmentation [7]; and MedCLIPSeg uses probabilistic cross-modal adaptation [4]. TextMatch studies multimodal consistency in semi-supervised segmentation [17], whereas ARSeg targets missing or imbalanced textual attributes [18]. These models motivate a controlled test of whether the training prompt still describes the transformed image.

Recent work also separates prompt content from fusion design. DescriptorMedSAM evaluates a structured progression from target name to name-shape-location descriptors and finds that shape and location provide complementary guidance [19]. Spatial-Aware Symmetric Alignment encodes directional constraints as region-level guidance while aligning descriptive and diagnostic expressions [20]. NTP-MRISeg instead casts image, text, and mask tokens as one autoregressive sequence [21]. These architecture- and descriptor-level advances are complementary to MEDEQUISEG, which asks both whether a supplied spatial descriptor remains valid after training-time geometry and whether the trained prediction remains case-specifically text dependent.

1.1.2 Image-only reference models

U-Net [8], U-Net++ [9], nnU-Net [11], Rolling-Unet [10], U-Mamba [12], and U-KAN [13] demonstrate that a strong visual pathway can perform well without text. A multimodal method should therefore be compared against both text-guided baselines and competitive image-only models. RollingUNet, U-Net, U-Net++, nnU-Net, U-Mamba, and U-KAN are complete for all datasets and all three seeds under Protocol V3. This breadth is important because the strongest image-only reference differs across datasets.

1.1.3 Consistency under paired augmentation

Data augmentation generally assumes that the target remains valid after a transformation [14]. For image-text learning, this assumption extends to language. Yi et al. update predefined caption keywords when horizontal flipping image-caption pairs and combine this with semantics-preserving appearance augmentation [15]. In multimodal medical segmentation, RBTCA instead grounds textual cues in image regions before augmenting a modality-aware representation [16]. Chai et al. identify the same flip/rotation conflict and avoid it by fusing text into a pseudo-image before augmentation [22]. Semi-MedRef synchronizes positional spans during patch mixing and perturbs position phrases in a semi-supervised teacher-student setting [23]. MEDEQUI-SEG therefore does not claim priority for the general alignment problem, paired caption augmentation, or exchanging individual left/right tokens. Its specific contribution is the cross-dataset evidence chain: an explicit lexical map for a finite image-coordinate vocabulary, deterministic exact-plan controls, a complete ordered ablation, and a privileged-prompt reliability audit.

2 Methods

2.1 Dataset and prompt provenance

Table 1 reports the locked manifests. The image and mask sources are BUSI [25], CVC-ClinicDB [26], BUS-BRA [27], BRISC [28], and QaTa-COV19 [29]. Their paired prompts are distributed by the MedCLIPSeg repository [4]. For lesion cases, MedCLIPSeg extracts class, coarse location, component count, shape, and tone from the ground-truth mask or label and fills language-model-generated templates. Thus these are target-derived synthetic referring captions, not prospective clinical reports. This benchmark construction is shared by the image+text methods in Table 7, but it gives them target-derived information unavailable to image-only methods and limits both clinical interpretation and cross-modality comparison.

2.2 Prompt-evidence scope and real-text validation

The public MedCLIPSeg mask/label-derived captions are restricted to mechanism analysis. They do not constitute real-text validation. Deployment-oriented evidence, which is not available in the present study, would require blinded clinician-authored or clinician-verified prompts generated from images without access to masks. Such a study must freeze the vocabulary and annotation instructions before opening the

Table 1 Locked image-level split and realized training-prompt audit. Unique and spatial coverage are computed on training prompts. Realized change enumerates all case-epoch events under the frozen follow-up MEDEQUISEG plan for 100 epochs and reports the mean $\pm$ sample standard deviation across seeds 123, 456, and 789. Public prompts are target-derived synthetic captions rather than clinical text.

Dataset	Train	Val.	Test	Prompt	Unique	Spatial (%)	Change (%)
BUSI	624	78	78	Target-derived	198	82.85	20.27 ± 0.10
ClinicDB	490	61	61	Target-derived	202	100.00	28.48 ± 0.21
BUS-BRA	1311	282	282	Target-derived	75	100.00	12.44 ± 0.12
BRISC	3881	974	892	Target-derived	359	78.23	5.75 ± 0.04
COVID-19	5716	1429	2113	Target-derived	269	100.00	45.75 ± 0.06

untouched test set and must compare correct, shuffled, fixed, and empty prompts. We do not use machine-generated captions as a substitute for clinical text evidence.

Spatial coverage counts prompts containing any member of the finite dictionary, including center, middle, top, bottom, upper, lower, left, and right compounds. This count alone overstates intervention frequency because central phrases can be invariant and geometric operations are sampled probabilistically. We therefore replay the exact keyed random generator for every training case, epoch, and seed and report the realized proportion of events whose final prompt string differs from the original. Across 100 epochs and seeds 123, 456, and 789, the rates range from $5.75 \pm 0.04\%$ for BRISC to $45.75 \pm 0.06\%$ for COVID-19. Complete event counts and operation rates are reported in Supplementary Table S7.

2.3 Protocol V3

The protocol identifier is `MEDSEG.TEXT.V3_20260710`. Manifests are fixed across seeds. Seeds 123, 456, and 789 change parameter initialization, minibatch order, and augmentation sampling; they do not resplit the data. Validation selects the best checkpoint, and the test split is accessed only by the final prediction and evaluation scripts. Predictions are restored to the native ground-truth resolution before scoring.

The protocol hash locks the dataset registry, manifests, split roles, mask rules, seeds, threshold, and evaluation specification, but it does not encode the full training policy or evaluator source. In particular, the original V3 LCAugv2 router restricted ClinicDB and COVID-19 to horizontal flips and excluded affine, translation, and zoom. The shared-plan rows in the 90-cell matrix instead use the separately dated R11 follow-up policy: all five public datasets receive the same multi-geometry, appearance, mild affine, translation, and zoom sampler. That policy was frozen before the paired R11/no-rewrite matrix, but it is an amendment on the fixed V3 data and evaluation protocol rather than part of the original protocol lock. Accordingly, we use the V3 protocol hash only for data/evaluation identity and retain a separate run-configuration hash, code hash, manifest hash, cache hash, checkpoint hash, and original command log for each training task. We do not describe the shared-plan matrix as prospectively confirmatory under the original July 10 augmentation specification (Supplementary Table S11).

BRISC contains duplicated images with conflicting masks. Protocol V3 drops conflicting duplicate groups while preserving exact holdout membership; this sanitation rule is part of the dataset registry and protocol hash. The public MedCLIPSeg manifests inherit the repository’s Train/Val/Test folders and do not expose a populated patient- or sequence-ID field; consequently, we describe these as fixed image-level holdouts rather than claiming patient-level separation. A post hoc filename audit can recover subject IDs for part of COVID-19: no recovered subject crosses into the fixed test set, but 434 subjects occur in both its original training and validation folders. We therefore performed a separately hashed sensitivity rerun that assigns every recoverable subject to only one training or validation split while preserving the fixed test set. Rows without recoverable IDs remain case-level groups. The grouped-validation training results are reported in Supplementary Table S10, and the reported-checkpoint test-set cluster sensitivity additionally addresses uncertainty from repeated test images.

We report Dice and IoU in the main table and additionally export precision, recall, normalized surface Dice (NSD) at two pixels, HD95, and ASSD. A both-empty case receives perfect overlap and zero distance. A one-sided empty case receives zero overlap and infinite distance, and the number of such boundary failures is reported with the distance metrics. Supplementary Tables S1–S5 report NSD, finite-case HD95 and ASSD, and the one-empty failure rate for every method and dataset. Finite distance means are interpreted only together with the failure rate.

2.4 MedEquiSeg framework

2.4.1 Framework overview and reliability scope

MED EQUISEG comprises a visual segmentation pathway, a frozen biomedical text representation pathway, a shared transformation plan, and an evaluation-time semantic audit (Fig. 1). The visual pathway fine-tunes the CausalCLIPSeg RN50 encoder and retains its scale-wise supportive/inferior feature partition, and two text-conditioned projection heads. Cached 512-dimensional pooled BioMedCLIP features pass through a trainable state projection and condition both heads; each evaluated head contains one active ATConv operator. During training, the shared plan keeps the image and mask geometrically aligned and updates discrete spatial phrases when their image-frame meaning changes. During evaluation, correct, shuffled, fixed, and empty prompts test whether the checkpoint responds to case-specific text. We use *prompt-reliability-aware* in this operational sense: the framework protects and audits prompt reliability. It does not estimate per-case uncertainty or contain a learned reliability gate.

2.4.2 Frozen follow-up shared plan

The shared rows of the ordered ablation use the R11 follow-up policy defined above. For case identifier c , epoch e , and training seed s , a keyed random generator samples

$$A_{c,e,s} = S(H, V, R, F, Z, C, G, N \mid c, e, s), \quad (1)$$

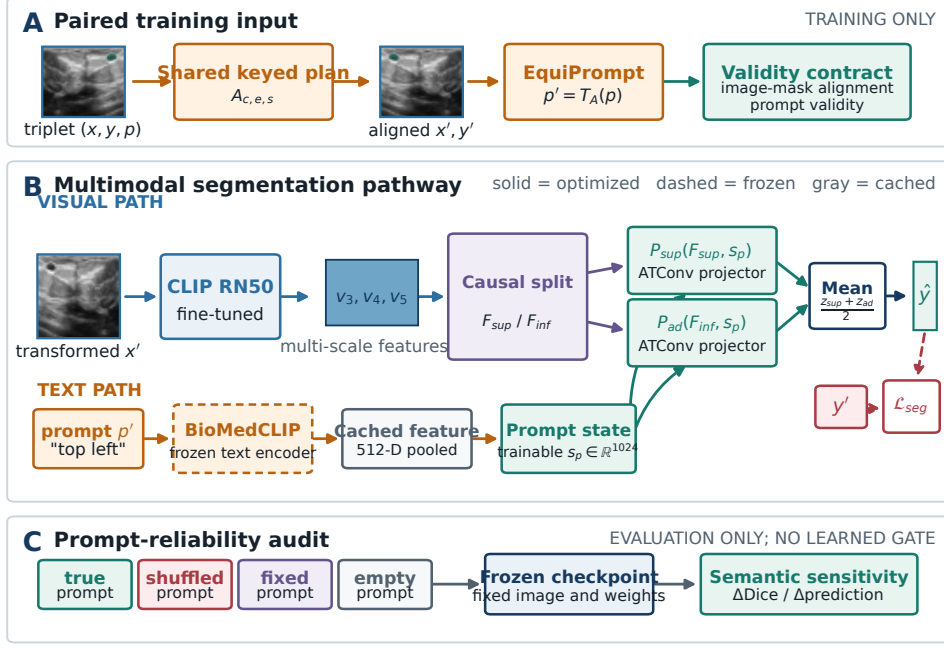


Fig. 1 Architecture of MEDEQUISEG. A public BUSI training case is used only to illustrate the paired transformation. (A) During training, a keyed online plan produces an aligned image–mask pair and the corresponding EquiPrompt lexical view. (B) The fine-tuned CLIP RN50 features are partitioned into supportive and inferior streams, where the latter is the complement of the supportive mask. A trainable state projection maps the cached frozen BioMedCLIP feature to s_p , which parameterizes the two causal projection heads; each head contains one active ATConv operator. Their logits are averaged before segmentation loss. The transformed target y' provides the second input to the BCE–Dice loss. Solid modules are optimized, dashed modules are frozen, and the gray feature is cached offline. (C) At evaluation, true, shuffled, fixed, and empty prompts audit semantic sensitivity of the same frozen checkpoint; this audit is not a learned reliability gate.

where H and V are horizontal and vertical flips, R is a 90-degree rotation, F is a mild affine transform with translation, Z is zoom, and C , G , and N denote color jitter, gamma, and Gaussian noise. The same realized spatial plan is applied to image x and mask y :

$$x' = A_{c,e,s}^x(x), \quad y' = A_{c,e,s}^y(y). \quad (2)$$

Images use bilinear interpolation, masks use nearest-neighbor interpolation, and the mask is never exposed to appearance operations. The training set is not physically duplicated; a new deterministic plan is sampled online each epoch.

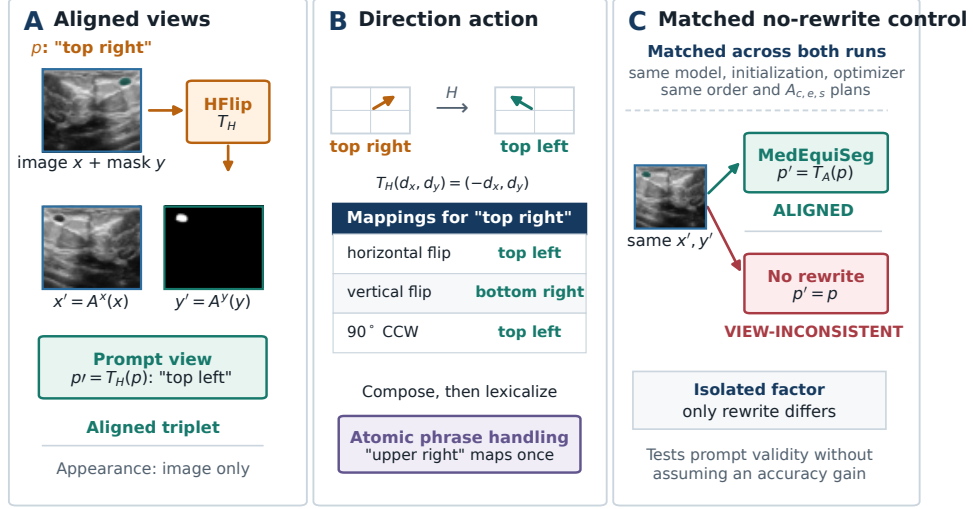


Fig. 2 Equiprompt synchronization and its exact mechanism control, illustrated with a public BUSI training case. (A) One realized transform creates aligned image, mask, and prompt views; appearance-only operations affect the image alone. (B) Direction phrases are represented in image coordinates, transformed compositionally, and lexicalized only after all discrete operations; multiword phrases are treated atomically. (C) MEDEQUISEG and MEDEQUISEG w/o Prompt Rewrite are separate runs matched in architecture, initialization, optimizer, data order, and realized augmentation sequence. Only the lexical map differs, isolating prompt validity from ordinary geometric augmentation without assuming that the rewrite must improve accuracy.

2.4.3 Compositional direction rewriting

Each recognized direction is mapped to a vector $d = (d_x, d_y)$ with components in $\{-1, 0, 1\}$. Horizontal and vertical flips apply

$$T_H(d_x, d_y) = (-d_x, d_y), \quad T_V(d_x, d_y) = (d_x, -d_y), \quad (3)$$

and a counter-clockwise 90-degree rotation applies

$$T_R(d_x, d_y) = (d_y, -d_x). \quad (4)$$

Sequential transforms compose these maps before converting the result back to the original lexical family. Thus, *upper right* remains an atomic target rather than two unrelated replacements. Comma-separated targets are parsed as separate atoms. The BioMedCLIP cache contains the finite lexical closure reachable under all compositions, preventing an augmented prompt from silently falling back to a missing embedding.

2.4.4 Augmentation profile and scope

MED EQUISEG uses one locked mild profile: $p_H = 0.35$, $p_V = 0.10$, and $p_R = 0.12$; when rotation is selected, 90° , 180° , and 270° are sampled uniformly. Mild affine is selected with probability 0.15, with angle in $[-5^\circ, 5^\circ]$ and translation in $\pm 2.5\%$ of image size. Zoom is selected with probability 0.25 in $[1.00, 1.07]$. Color jitter, gamma, and noise have probabilities 0.45, 0.25, and 0.25, respectively; jitter magnitude is 0.05, gamma is in $[0.90, 1.12]$, and noise standard deviation is 0.006 after normalization.

Only flips and right-angle rotations trigger exact direction rewriting. Small affine rotations, translation, and off-center zoom do not exchange coordinate axes, but they can move a target across a coarse location bin. Consequently, MED EQUISEG guarantees lexical consistency for the discrete axis-changing operations, not for arbitrary continuous spatial descriptions. The small ranges bound this approximation. We additionally ran MED EQUISEG + Location Recompute, a benchmark-only diagnostic that recomputes coarse position phrases from the connected-component centroids of the transformed training mask whenever affine, translation, or zoom is sampled. Although it uses the same transform distribution, model family, and training seeds, its augmentation seed key differs from the main run and the realized per-case transforms are therefore not exactly paired. We report it only as an exploratory same-distribution diagnostic.

2.4.5 Multimodal segmentation pathway

The evaluated model initializes the CausalCLIPSeg visual pathway from the official OpenAI CLIP RN50 checkpoint and fine-tunes it end to end. Prompts are represented by cached, frozen 512-dimensional pooled features from `microsoft/BiomedCLIP-PubMedBERT_256-vit_base_patch16_224`. A trainable linear projection maps each pooled feature to the 1024-dimensional state consumed by both CausalCLIPSeg projectors. Each projector converts this state into a sample-specific 3×3 kernel and bias. One same-channel convolution in each active projector is replaced by ATConv [24], giving two effective ATConv operators in the evaluated graph. The adapter also instantiates expanded token features, but all token-level fusion switches are disabled in the reported configuration; those tokens therefore do not enter the effective forward path. Plugin metadata records four substitutions because two additional replacements target dormant legacy FPN members that are not called by the released `FPN_AD.forward`; we exclude them from the architectural claim. A post-run forward-hook audit loaded the reported BUSI seed-123 checkpoint and passed one real test case through the model: both projector operators were called once with non-zero outputs, while the two registered neck candidates were never called (Supplementary Table S11). Training uses a BCE–Dice objective with Dice weight 0.5. Text conditions visual decoding but does not directly generate the target mask. MED EQUISEG w/o Prompt Rewrite has the identical architecture, objective, and sampled image–mask plans; only prompt rewriting is disabled.

2.5 Experimental design

2.5.1 Training and evaluation

Inputs are resized to 224×224 for training. All MEDEQUISEG runs use 100 epochs, batch size 8, AdamW with initial learning rate 10^{-4} and weight decay 10^{-4} , and cosine decay to 10^{-6} . Checkpoint selection uses validation Dice. Evaluation uses a fixed threshold of 0.5. Predictions are restored from the 224×224 inference grid to the native reference-mask grid before scoring. The principal MEDEQUISEG and image-baseline environment used Python 3.11.15 and PyTorch 2.5.1 with CUDA 12.1. The nnU-Net/MedCLIPSeg environment used nnU-Net v2.8.0 and MONAI 1.3.0; the official U-Mamba path used Python 3.10.20 and nnU-Net v2.1.1. Runs used NVIDIA RTX 4090 GPUs with driver 580.119.02. A package-level environment record lists the remaining core library versions.

Online complexity is measured from the frozen BUSI seed-123 checkpoints on one NVIDIA GeForce RTX 4090 with batch size 1, FP32, and 224×224 inputs. After model-specific warm-up, each forward pass is timed with wall-clock CUDA synchronization; data loading and mask post-processing are excluded. The MEDEQUISEG and LViT measurements include cached-text lookup but not the one-time offline BioMedCLIP or PubMedBERT encoding step. MedCLIPSeg instead runs its released online UniMedCLIP/BiomedBERT path and is measured with both one probabilistic sample and its official 30-sample test setting. Parameter counts include all modules instantiated by each checkpoint loader and are not pruned to output-connected subgraphs. Because the prompt-consistency policy is active only during training, it adds no inference branch or parameter beyond the reported MEDEQUISEG backbone.

The expression *same protocol* in this paper means the same locked manifest, split roles, held-out test set, mask sanitation, prediction restoration, and metric implementation. It does not imply that architectures with different official recipes share identical losses, input sizes, or augmentation. We therefore use the more precise phrase *same data and evaluation protocol* in comparative claims.

2.5.2 Baselines and controls

The completed comparison includes MedCLIPSeg, CausalCLIPSeg, LViT, RollingUNet, U-Net, U-Net++, nnU-Net, U-Mamba, and U-KAN [4, 6–13]. The first three and MEDEQUISEG are multimodal; the remaining six models are image-only. Every method has three optimization seeds on every dataset. U-Net, U-Net++, nnU-Net, U-Mamba, and U-KAN contribute 75 public dataset–model–seed cells; RollingUNet contributes 15 more, for 90 image-only public cells in total. The contextual CausalCLIPSeg row (internal run V3_R0) and the Base reference (V3_ABL_BASE) use the same CLIP-text/horizontal-flip recipe but are independent three-seed training matrices; the latter is the reference for the ordered component ablation, whereas the former is retained in the broad model table.

MEDEQUISEG w/o Prompt Rewrite is the primary mechanism control. Four inference-time semantic controls are also defined without retraining: correct case text, seed-fixed shuffled text, a training-derived fixed prompt, and an empty prompt. Correct text must exceed shuffled text in the Holm-adjusted paired rank screen before a

dataset enters the semantic-control visualization; a hierarchical interval excluding zero is additionally required for the evidence to be described as robust to seed uncertainty. The shuffled control applies a deterministic non-zero cyclic shift (control seed 123) to the ordered test prompts. It preserves the complete prompt multiset and assigns every row another row’s prompt, although duplicated prompt strings can remain textually unchanged; no class or attribute stratum is preserved. To assess this confounding directly, a post hoc frozen-checkpoint control permutes the complete prompt multiset separately within target-presence and diagnostic-class strata. Strata are constructed only from released case identifiers and prompt fields; the builder reads neither masks nor image pixels. Within each stratum, a deterministic assignment maximizes changed prompt strings while preserving one-to-one prompt closure. This changes 3,413 of 3,426 public test prompts; 13 BUSI normal cases remain unchanged because every available within-stratum prompt string is identical. Fixed and empty prompts are secondary descriptive controls. The released target-derived captions on the five public datasets define the formal but privileged referring benchmark. Additional prompts generated directly from a case’s mask for oracle diagnostic experiments are a separate category and are excluded from formal performance tables.

A frozen follow-up ordered component ablation uses the fixed V3 manifests and evaluator for the base CausalCLIPSeg pathway, BioMedCLIP substitution alone, ATConv alone, their combination, the shared deterministic plan without rewriting, and the complete MEDEQUISEG configuration. Every configuration contains all five public datasets and seeds 123, 456, and 789, yielding 90 unique dataset–configuration–seed cells. The first four configurations use the same hflip policy; the last two share the same richer online plan and differ only in whether direction phrases are rewritten. Historical engineering runs using other prompt-generation or evaluation definitions are excluded.

2.5.3 Endpoints and statistical analysis

The main accuracy-centered contrast reported in this retrospective ordered ablation is the complete MEDEQUISEG configuration minus the horizontal-flip Base reference in equal-dataset Public-5 macro Dice. This contrast was selected after the original frozen promotion gate and is therefore interpreted as post hoc rather than as a prospectively declared primary endpoint. The mechanism contrast frozen before completion of the paired matrix is MEDEQUISEG minus MEDEQUISEG w/o Prompt Rewrite under an exactly matched visual transformation plan. The ordered ablation additionally reports the incremental BioMedCLIP and ATConv configurations. Both contrasts use case-first paired inference after averaging the three seed values per case. Dataset-specific Dice and IoU differences and boundary measures are secondary endpoints. The primary semantic-control endpoint is true versus shuffled Dice; fixed and empty controls are descriptive. The presence/class-stratified derangement is a post hoc sensitivity analysis of the true–shuffled contrast, not a replacement primary endpoint. It uses the same case-first three-seed reduction, 10,000 paired bootstrap replicates, and five-dataset Holm family as the other semantic comparisons. Cross-model and cross-modality comparisons in Table 7 are contextual rather than confirmatory because architectures and input information differ.

Two additional training controls were completed after the 90-cell matrix. A constant-text run keeps the complete recipe and shared plan but replaces every cached prompt feature with the same empty-prompt embedding throughout training and test. A Base-plus-plan run applies the richer shared plan to the Base-reference recipe without BioMedCLIP substitution, ATConv, or rewriting. Their comparisons with the earlier matrix are post hoc sensitivity analyses because the runner code snapshot changed between batches, although manifests, evaluator, epochs, optimizer settings, and command logs are retained. We therefore use these controls to assess attribution, not as new confirmatory endpoints.

Seed-level results are reported as mean and sample standard deviation. For hypothesis tests, per-case metrics are first averaged across seeds for each method, then compared with paired Wilcoxon signed-rank tests; paired t -tests are reported as sensitivity analyses. Family-wise multiplicity is controlled with Holm correction [30]. Confidence intervals use 10,000 paired-case bootstrap replicates after the three seed values for each case have been averaged. For the Public-5 macro effect, resampling is stratified within each dataset before the five dataset-level effects are averaged, so each benchmark receives equal weight. Training-seed variability is retained in the reported mean and sample standard deviation but is not treated as an additional set of independent cases. Bootstrap intervals estimate the mean effect, whereas the Wilcoxon test targets paired rank differences; the two summaries are interpreted together rather than treated as interchangeable significance decisions. Case-level tests use manifest rows as the paired units and are interpreted subject to the patient-grouping limitation described below. Because the COVID-19 test set contains repeated images from 474 recoverable subjects, a post hoc sensitivity analysis retains the image-weighted point estimate but resamples subjects and training seeds for the interval and applies paired tests to equal-subject mean differences. This clustered analysis is reported separately and does not replace the pre-specified image-level analysis.

2.5.4 Software-assisted manuscript preparation

OpenAI Codex was used for code-assisted reproducibility checks and English language editing. The authors inspected the generated changes and retained full responsibility for the experimental code, numerical results, citations, and manuscript text.

3 Results

3.1 Segmentation accuracy and component contributions

Across the five public datasets, MEDEQUISEG attains equal-dataset macro Dice of 87.87% and IoU of 80.97%. This is 1.34 Dice pp and 1.65 IoU pp above the next reproduced image+text model, CausalCLIPSeg, and 2.91 Dice pp and 2.74 IoU pp above the highest-macro image-only architecture, nnU-Net. These macro comparisons summarize the fixed public benchmark panel; the cross-modality difference remains descriptive because image+text methods receive target-derived captions. Per-dataset results are reported below to expose heterogeneity rather than treating the macro value as a universal ranking.

Table 2 Equal-dataset macro segmentation accuracy on the five public benchmarks. Values are the arithmetic mean of the three-seed dataset means in Table 7. Bold and underline rank models within modality. Cross-modality differences are descriptive because the image+text models receive target-derived synthetic captions unavailable to the image-only models.

Method	Dice (%)	IoU (%)
<i>Image+text methods</i>		
MedEquiSeg (Ours)	87.87	80.97
CausalCLIPSeg Baseline	<u>86.53</u>	<u>79.32</u>
MedCLIPSeg	86.11	78.27
LViT	84.85	77.37
<i>Image-only contextual references</i>		
nnU-Net	84.96	78.23
U-Mamba	<u>84.73</u>	<u>77.94</u>
U-KAN	84.38	77.02
U-Net++	84.10	76.65
RollingUNet	82.40	74.62
U-Net	79.62	71.20

Table 3 Frozen follow-up three-seed ordered component ablation on the five public datasets (6 configurations $\times$ 5 datasets $\times$ 3 seeds = 90 cells). Values are equal-dataset Public-5 Dice and IoU percentages (mean $\pm$ sample standard deviation). BMC denotes pooled BioMedCLIP conditioning; ATConv denotes the two forward-active projector replacements; Plan denotes the R11 shared training policy applied on the fixed V3 data and evaluation protocol. Bold and underline identify the best and second-best configurations. The complete per-dataset matrix is reported in Supplementary Table S10.

Configuration	BMC	ATConv	Plan	Rewrite	Dice	IoU
Base model	–	–	–	–	86.41 $\pm$ 0.35	79.22 $\pm$ 0.34
+ BioMedCLIP	✓	–	–	–	86.67 $\pm$ 0.08	79.52 $\pm$ 0.06
+ ATConv	–	✓	–	–	86.63 $\pm$ 0.32	79.42 $\pm$ 0.31
+ BioMedCLIP + ATConv	✓	✓	–	–	87.03 $\pm$ 0.15	79.85 $\pm$ 0.23
+ Shared plan (no rewrite)	✓	✓	✓	–	87.96 $\pm$ 0.05	81.10 $\pm$ 0.08
MedEquiSeg (Ours)	✓	✓	✓	✓	<u>87.87 $\pm$ 0.17</u>	<u>80.97 $\pm$ 0.13</u>

The frozen follow-up ordered component ablation contains all 90 planned cells. Public-5 macro Dice rises from 86.41 $\pm$ 0.35% for the base pathway to 86.67 $\pm$ 0.08% with BioMedCLIP, 86.63 $\pm$ 0.32% with ATConv, and 87.03 $\pm$ 0.15% with both components. Adding the shared deterministic plan without rewriting gives 87.96 $\pm$ 0.05%; the complete MEDEQUISEG configuration gives 87.87 $\pm$ 0.17%. The corresponding full-framework minus base-model differences are +1.46 Dice pp (95% dataset-stratified bootstrap CI +0.95 to +1.99) and +1.74 IoU pp (95% CI +1.13 to +2.34). Dataset-level Dice point estimates are positive throughout; only the ClinicDB interval includes

zero. The component increments are descriptive because this ordered ablation does not estimate every higher-order interaction, but it shows that neither BioMedCLIP nor ATConv alone accounts for the final improvement.

Table 4 Attribution-oriented Public-5 macro contrasts. Deltas are percentage points with 95% dataset-stratified paired-case bootstrap confidence intervals after averaging each case across the three training seeds. The first two rows belong to the same 90-cell code snapshot; the remaining rows are post hoc sensitivity controls completed after that matrix and are not treated as prospectively controlled endpoints.

Comparison	Dice Δ [95% CI]	IoU Δ [95% CI]
MED EQUISEG – Base reference	+1.46 [+0.95, +1.99]	+1.74 [+1.13, +2.34]
MED EQUISEG – w/o Rewrite	−0.09 [−0.30, +0.10]	−0.13 [−0.39, +0.10]
Base + shared plan – Base reference	+1.42 [+0.81, +2.13]	+1.70 [+1.05, +2.40]
MED EQUISEG – Base + shared plan	+0.04 [−0.40, +0.41]	+0.05 [−0.39, +0.45]
MED EQUISEG – constant-text full model	+1.90 [+1.16, +2.75]	+2.10 [+1.34, +2.98]

The follow-up controls further localize the aggregate effect. Base plus the shared plan reaches $87.83 \pm 0.23\%$ Dice and differs from the Base reference by +1.42 pp (95% CI +0.81 to +2.13). The complete model differs from Base plus the shared plan by only +0.04 pp (95% CI −0.40 to +0.41), providing no aggregate evidence for an additional contribution from the combined BioMedCLIP–ATConv–rewrite stack beyond the plan. Conversely, the complete model differs from its constant-text training control by +1.90 pp (95% CI +1.16 to +2.75), with intervals excluding zero on BUSI and COVID-19 but not on ClinicDB, BUS-BRA, or BRISC. Thus case-specific privileged text matters in aggregate, but its evidence is concentrated rather than uniform.

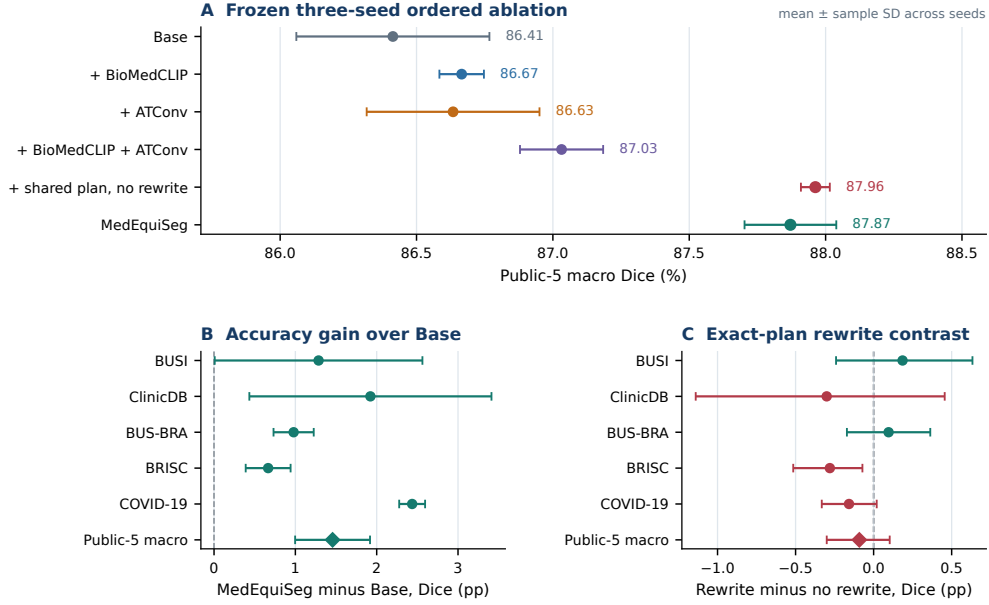


Fig. 3 Accuracy-centered follow-up ordered component ablation on the fixed V3 data and evaluation protocol. (A) Public-5 macro Dice for all six configurations; points are three-seed means and bars are sample standard deviations. (B) Paired three-seed Dice differences between MEDEQUISEG and the Base pathway for each public dataset and their equal-dataset macro; bars are sample standard deviations. (C) Dice differences between MEDEQUISEG and the shared-plan no-rewrite control. Dataset bars and the macro diamond are 95% paired-case bootstrap intervals after three-seed case averaging. The Public-5 macro difference from Base is positive, whereas the rewrite-only contrast is centered near zero.

3.1.1 Matched rewrite and semantic-control evidence

Across three seeds, exact-plan rewriting changes Public-5 macro Dice by -0.09 pp (95% dataset-stratified bootstrap CI -0.30 to 0.10 pp) and macro IoU by -0.13 pp (95% CI -0.39 to 0.10 pp). Dataset-level Dice differences range from $+0.19$ pp on BUSI to -0.30 pp on ClinicDB. BRISC and COVID-19 have Holm-adjusted Wilcoxon support for small negative rank shifts; the paired-case bootstrap interval excludes zero for BRISC but not COVID-19. The macro interval spans zero. Taken together, the exact control provides no evidence that prompt rewriting improves average segmentation accuracy and identifies a small adverse BRISC shift under the executed policy.

Trigger frequency helps interpret these differences but does not explain them alone. COVID-19 has the highest realized prompt-change rate ($45.75 \pm 0.06\%$) yet its rewrite effect is slightly negative. BUSI has a lower change rate ($20.27 \pm 0.10\%$) and a small positive point estimate. Trigger frequency alone therefore does not predict the accuracy contrast; prompt specificity, visual semantics, and optimization interactions remain plausible moderators.

The semantic controls further separate prompt use from rewrite benefit. True minus shuffled Dice ranges from $+0.77$ pp on BUS-BRA and $+1.88$ pp on ClinicDB to $+39.05$

Table 5 Strict three-seed exact-plan rewrite control on the five public datasets. MEDEQUISEG and MEDEQUISEG w/o Prompt Rewrite receive identical online image-mask transforms; only MEDEQUISEG rewrites directional text. Method values are mean $\pm$ sample standard deviation. Deltas and 95% confidence intervals are percentage points from a paired-case bootstrap after averaging each case across the three training seeds. $p_{W,H}$ and $p_{t,H}$ are two-sided Wilcoxon and paired t -test p -values with Holm correction across the five datasets, separately by metric.

Dataset	Metric (%)	MEDEQUISEG	w/o Rewrite	Δ pp [95% CI]	$p_{W,H}$	$p_{t,H}$
BUSI	Dice	85.78 $\pm$ 0.45	85.60 $\pm$ 0.58	+0.19 [−0.24, 0.63]	.640	1.000
	IoU	79.49 $\pm$ 0.23	79.36 $\pm$ 0.63	+0.13 [−0.41, 0.70]	.805	1.000
ClinicDB	Dice	92.07 $\pm$ 0.71	92.37 $\pm$ 0.93	−0.30 [−1.14, 0.46]	.184	1.000
	IoU	86.36 $\pm$ 0.61	86.82 $\pm$ 1.11	−0.46 [−1.47, 0.50]	.182	1.000
BUS-BRA	Dice	90.24 $\pm$ 0.14	90.14 $\pm$ 0.16	+0.10 [−0.17, 0.36]	.184	1.000
	IoU	83.15 $\pm$ 0.10	83.03 $\pm$ 0.18	+0.13 [−0.20, 0.45]	.182	1.000
BRISC	Dice	87.18 $\pm$ 0.14	87.46 $\pm$ 0.20	−0.28 [−0.51, −0.07]	< .001	.076
	IoU	80.32 $\pm$ 0.13	80.59 $\pm$ 0.20	−0.27 [−0.50, −0.05]	< .001	.076
COVID-19	Dice	84.08 $\pm$ 0.13	84.24 $\pm$ 0.13	−0.16 [−0.33, 0.02]	.003	.315
	IoU	75.51 $\pm$ 0.14	75.69 $\pm$ 0.13	−0.18 [−0.38, 0.01]	.005	.294
Public-5	Dice	87.87 $\pm$ 0.17	87.96 $\pm$ 0.05	−0.09 [−0.30, 0.10]	—	—
	IoU	80.97 $\pm$ 0.13	81.10 $\pm$ 0.08	−0.13 [−0.39, 0.10]	—	—

Table 6 Strict inference-time semantic controls for trained MEDEQUISEG checkpoints on the five public datasets. No model is retrained; only the test prompt changes. Dice is mean $\pm$ sample standard deviation over three seeds. The last column is true minus shuffled Dice with a paired-case bootstrap 95% interval after three-seed averaging. Holm-adjusted Wilcoxon tests and the bootstrap intervals support a positive true-shuffled difference on all five datasets.

Dataset	True Dice	Shuffled Dice	Fixed Dice	Empty Dice	Δ pp [95% CI]
BUSI	85.78 $\pm$ 0.45	46.73 $\pm$ 1.41	16.69 $\pm$ 0.02	13.22 $\pm$ 5.35	+39.05 [29.85, 48.49]
ClinicDB	92.07 $\pm$ 0.71	90.20 $\pm$ 0.94	88.31 $\pm$ 1.31	78.71 $\pm$ 7.93	+1.88 [0.07, 4.22]
BUS-BRA	90.24 $\pm$ 0.14	89.47 $\pm$ 0.23	88.87 $\pm$ 0.28	68.22 $\pm$ 8.44	+0.77 [0.45, 1.15]
BRISC	87.18 $\pm$ 0.14	70.89 $\pm$ 2.12	75.31 $\pm$ 1.97	58.89 $\pm$ 5.30	+16.29 [14.26, 18.43]
COVID-19	84.08 $\pm$ 0.13	68.80 $\pm$ 0.82	68.54 $\pm$ 0.16	55.81 $\pm$ 3.82	+15.29 [14.49, 16.10]

pp on BUSI. Holm-adjusted Wilcoxon tests support a true-shuffled difference on all five public datasets, and the paired-case bootstrap intervals are positive throughout; ClinicDB has the smallest interval (+0.07 to +4.22 pp). This demonstrates sensitivity to privileged case-matched caption identity rather than clinical-report understanding or specifically spatial grounding, because the public captions contain target-derived attributes.

Presence/class matching substantially attenuates but does not eliminate this effect. True text exceeds the stratified derangement by +5.63 Dice pp in the equal-dataset Public-5 macro (95% CI +4.71 to +6.67), compared with +14.66 pp under unstratified shuffling. Four of five dataset bootstrap intervals are positive; Holm-adjusted rank support remains for BUSI, BRISC, and COVID-19, whereas BUS-BRA is neutral. Restricting the analysis to the 3,413 cases whose text string actually changes gives

+5.87 pp (95% CI +4.87 to +7.03). For the 474 recoverable COVID-19 subjects, equal-subject averaging gives +22.42 pp with a subject-cluster interval of +21.12 to +23.80. These post hoc results show that coarse class/presence mismatch explains a large share of the original shuffled gap while leaving a heterogeneous residual association with target-derived prompt identity (Supplementary Table S13).

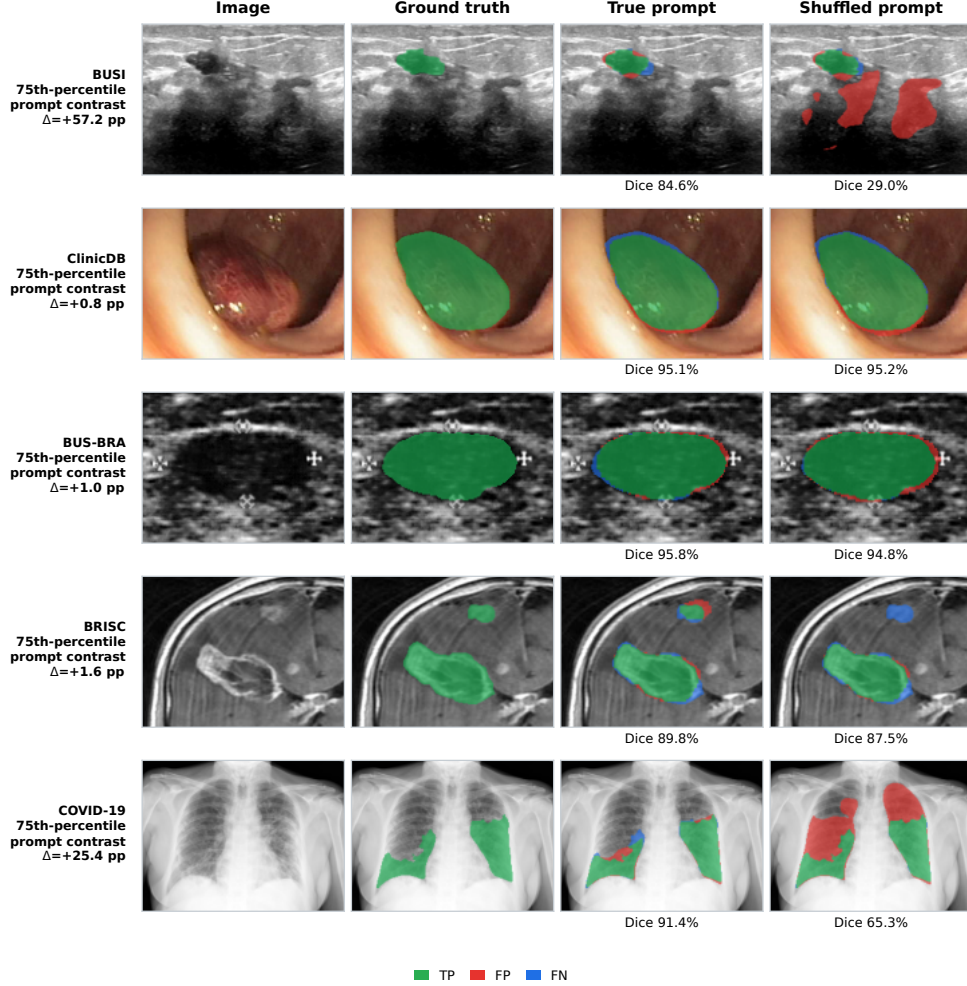


Fig. 4 Fixed-checkpoint true-versus-shuffled prompt audit. One non-empty case nearest the 75th percentile of the three-seed per-case Dice difference is shown for each dataset that passes the Holm-adjusted rank screen. Predictions are three-seed majority votes. The left-hand selection delta is the mean paired per-case difference across seeds and therefore need not equal the difference between the displayed majority-vote Dice values. Each row uses one shared context crop determined by the union of the ground truth and both predictions. Green denotes true positive, red false positive, and blue false negative. Selection was deterministic; exact prompts, case identifiers, and uncropped paths are archived in the selection CSV.

3.2 Per-dataset comparisons and qualitative error analysis

The broad model table is contextual. MEDEQUISEG has a higher mean Dice than reproduced MedCLIPSeg on ClinicDB (+0.11 pp), BUS-BRA (+1.77 pp), BRISC (+5.82 pp), and COVID-19 (+1.46 pp), and a lower mean on BUSI (−0.35 pp). Relative to the strongest image-only mean selected within each dataset, MEDEQUISEG is higher on BUSI (+8.68 Dice pp) and COVID-19 (+4.08 pp), and lower by 0.10–1.55 pp on the other public datasets. These cross-modality differences are descriptive, not evidence that privileged target-derived text yields a fair deployment advantage.

The COVID-19 grouped train-validation sensitivity preserved the fixed test set and removed all overlap among the 2,130 recoverable subject tokens. In this separate sensitivity checkpoint set, mean Dice was $83.99 \pm 0.07\%$ for the MEDEQUISEG configuration, compared with $84.18 \pm 0.13\%$ for its configuration-matched original-split rerun, a change of −0.19 pp. The corresponding no-rewrite and U-Net++ changes were +0.00 and +0.14 pp. Within the grouped-validation checkpoint set, rewriting is −0.29 Dice pp below no-rewrite (paired-case 95% CI −0.55 to −0.03); the equal-subject effect is −0.40 pp with a subject-cluster interval of −0.58 to −0.02 and Wilcoxon $p < .001$. Thus, the known recoverable-subject overlap has little effect on the comparison with U-Net++, while the grouped sensitivity reinforces rather than reverses the absence of a rewrite benefit. However, rows without recoverable subject tokens cannot be certified as patient grouped (Supplementary Table S10).

The mask-presence analysis (Supplementary Table S8) shows that BUSI contains 65 non-empty and 13 empty test masks. MEDEQUISEG and U-KAN have non-empty Dice values of $82.94 \pm 0.55\%$ and $79.70 \pm 0.48\%$, a difference of +3.24 pp, whereas the overall difference is +8.68 pp. Empty specificity is 100.00% for MEDEQUISEG and $64.10 \pm 17.76\%$ for U-KAN. Empty-mask cases therefore amplify, but do not fully account for, the BUSI difference. BRISC has the only other mixed test split (753 non-empty and 139 empty masks); the complete subgroup audit is reported rather than using empty cases as implicit evidence of lesion quality.

The canonical boundary results in Supplementary Tables S1–S5 follow the same dataset-dependent pattern. Because one-sided empty cases have infinite distance, finite HD95 and ASSD are interpreted only with the failure rate.

MEDEQUISEG instantiates 159.78 million parameters and requires 15.03 ± 0.82 ms per image. The training-time policy adds no inference parameters, and MEDEQUISEG w/o Prompt Rewrite has the same inference graph. These measurements describe implementation cost rather than an architecture-controlled speed comparison.

The pre-specified upper- and lower-tail examples in Figure 5 make the claim boundary visible: selected BUSI and COVID-19 cases favor MEDEQUISEG, whereas the ClinicDB and BUS-BRA cases expose false-positive or false-negative failure modes.

Table 8 Instantiated model size and synchronized single-image inference cost. Parameter columns count all modules stored in each instantiated model; they are not pruned to output-connected subgraphs. Requires-grad counts follow the checkpoint implementation. All timing measurements use batch size 1, FP32, 224×224 input, and an NVIDIA GeForce RTX 4090. Values are mean $\pm$ standard deviation after warm-up; data loading and post-processing are excluded. Cached text lookup is included for MEDEQUISEG and LViT, whereas MedCLIPSeg runs its online UniMedCLIP/BiomedBERT towers. S is the number of probabilistic samples requested from the released MedCLIPSeg forward path. A dagger marks image-only methods.

Method	Params (M)	Trainable (M)	ms/image	Images/s	Peak MB
U-Net++ [†]	2.41	2.41	6.80 $\pm$ 0.09	146.95	123
RollingUNet [†]	1.78	1.78	26.40 $\pm$ 0.65	37.88	31
LViT	39.93	39.93	11.35 $\pm$ 0.58	88.08	278
CausalCLIPSeg Baseline	165.26	165.26	18.83 $\pm$ 0.53	53.10	895
MedCLIPSeg ($S = 1$)	214.58	18.68	55.44 $\pm$ 6.97	18.04	1814
MedCLIPSeg ($S = 30$)	214.58	18.68	816.41 $\pm$ 12.91	1.22	1825
MedEquiSeg (Ours)	159.78	159.78	15.03 $\pm$ 0.82	66.55	874

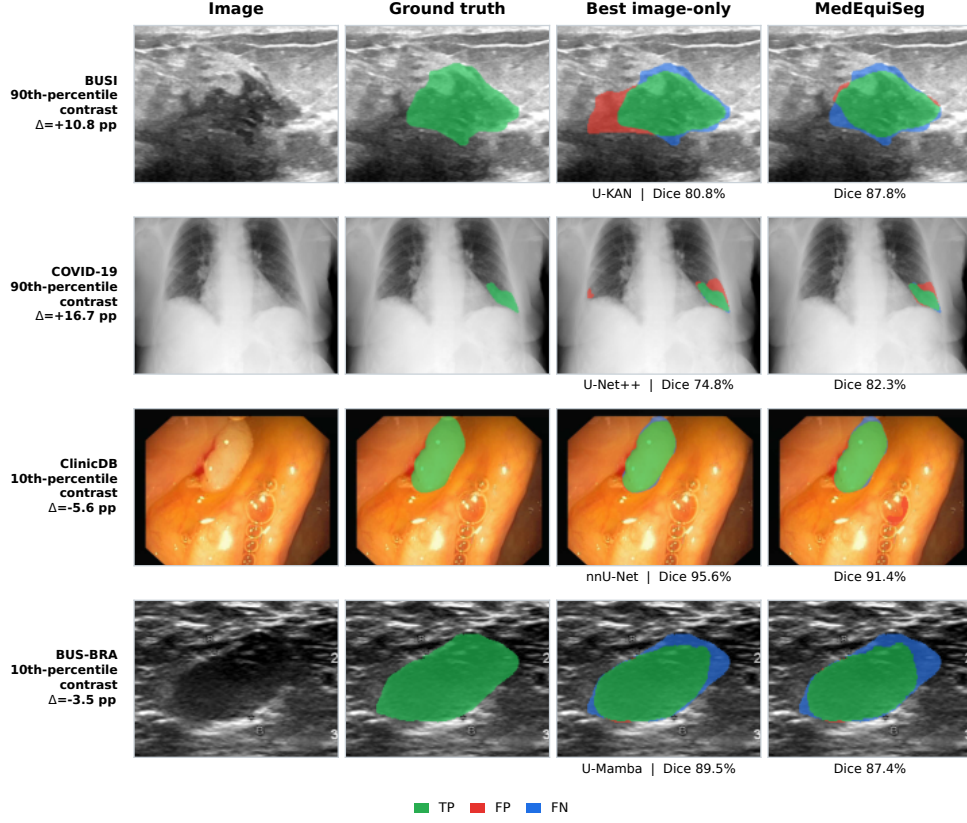


Fig. 5 Frozen-rule qualitative audit. BUSI and COVID-19 cases are nearest the pre-specified 90th percentile of the three-seed MEDEQUISEG minus image-only Dice difference; ClinicDB and BUS-BRA cases are nearest the 10th percentile. Predictions are pixel-wise majority votes. The left-hand selection delta is the mean paired per-case difference across seeds and therefore need not equal the difference between the displayed majority-vote Dice values. Green is true positive, red false positive, and blue false negative. Within each row, all columns use one context crop determined by the union of the ground truth and both predictions. The deterministic case and prompt record is provided in the results archive; examples are not additional tests.

4 Discussion

The principal empirical result is higher accuracy for the complete configuration than for the base multimodal pathway: +1.46 Dice pp (95% CI +0.95 to +1.99) and +1.74 IoU pp (95% CI +1.13 to +2.34) in the Public-5 ordered ablation. Because the contrast was selected retrospectively and the shared policy was a follow-up amendment on the fixed V3 manifests and evaluator, it supports an association on this benchmark panel rather than a prospectively confirmed universal effect. MEDEQUISEG also has the highest equal-dataset macro Dice and IoU among the reproduced models, but this does not mean that it is the best method on every dataset. The strongest image-only result is higher on ClinicDB, BUS-BRA, and BRISC, whereas MEDEQUISEG is higher on BUSI and COVID-19. Reporting both macro and per-dataset results is therefore necessary: the macro captures consistency across the benchmark panel, while the dataset table exposes modality-specific failure and competitive margins.

The component study localizes this gain. BioMedCLIP and ATConv individually change the Public-5 macro by only +0.26 and +0.22 Dice pp relative to the base pathway; using both gives +0.62 pp. The shared plan without rewriting reaches +1.55 pp, while the complete configuration reaches +1.46 pp. The post hoc Base-plus-plan control independently reaches +1.42 pp, while the complete-model residual over that control is +0.04 pp (95% CI −0.40 to +0.41). The shared policy therefore explains nearly all of the observed macro point improvement; the data do not establish an additional aggregate accuracy benefit from the complete BioMedCLIP–ATConv–rewrite stack beyond that policy. The runtime audit confirms that ATConv was not globally disabled: two projector operators execute and produce non-zero outputs. It also confirms that the other two registered candidates are dormant, so these results support only the effective two-projector adapter and not an ATConv4 claim.

Prompt reliability remains an interpretive control rather than the primary performance claim. A shared plan guarantees image–mask alignment; the language map addresses prompt validity under a subset of spatial transforms; and semantic controls test whether the checkpoint depends on case-matched text. The exact-plan control is essential because comparison with an unaugmented baseline cannot distinguish language consistency from ordinary regularization. Likewise, segmentation accuracy without shuffled or empty prompts cannot establish reliable text use.

The semantic-control pattern constrains interpretation. Large true–shuffled gaps on all five public sets show that prompt identity affects the prediction. After matching target presence and diagnostic class, the Public-5 gap falls from +14.66 to +5.63 Dice pp. This attenuation identifies coarse class/presence confounding in the unstratified control, while the remaining effect is concentrated in BUSI and COVID-19 and is nearly absent on BUS-BRA. The residual therefore supports sensitivity to privileged case-specific prompt content, not a general claim of language grounding. The constant-text training control also differs from the complete model by +1.90 Dice pp in the Public-5 macro, but only BUSI and COVID-19 have dataset-level intervals excluding zero. Target-derived captions therefore make this a concentrated privileged synthetic referring effect rather than clinical-report understanding.

The result is consistent with the broader paired-augmentation literature rather than a priority claim. Augment the Pairs rewrites predefined caption keywords during

horizontal flips, while RBTCA builds a region-grounded modality-aware representation before augmentation [15, 16]. Our added evidence is chiefly diagnostic: a deterministic exact-plan no-rewrite comparison shows that semantic correction need not improve overlap accuracy even when the shared augmentation policy does.

The prompt-design evidence from DescriptorMedSAM and the explicit spatial reasoning in SSA reinforce that location language can be useful when it is both informative and correctly grounded [19, 20]. NTP-MRISeg further shows that text can be integrated through a unified generative objective rather than a conditional decoder [21]. Their published values are not inserted into our quantitative tables because their datasets, prompt construction, splits, architectures, and evaluation recipes are not identical to Protocol V3. Accordingly, they contextualize the design space but do not serve as same-protocol evidence for or against MEDEQUISEG.

The BRISC result requires particular care because duplicate sanitation changes the effective sample count. The sanitation rule is fixed before training and applied to all compared methods.

4.1 Limitations and next evidence

First, the V3 hash identifies fixed data and evaluation choices, not the complete training policy. The richer R11 shared plan changes the original dataset routing and adds affine, translation, and zoom, so the 90-cell matrix is a frozen follow-up ablation rather than a strict execution of the original July 10 augmentation specification. Existing run-configuration and code hashes preserve the identity of the executed batch, but the recorded Git commit omits several orchestration/evaluation files and the currently available release copies do not reproduce the aggregate execution-code digest. Exact clean-clone reconstruction of the 90-cell source snapshot is therefore not demonstrated. The original completion digest also omitted command-line fields such as batch size and worker count; archived command logs are needed to recover them. Second, four ATConv modules are present in checkpoint metadata, but only the two projector modules enter the released forward graph. The reported parameter count includes the two dormant neck modules, and no ATConv4 claim is made. Third, the exact three-seed rewrite control shows small heterogeneous effects and a macro bootstrap interval spanning zero. BRISC has a small negative paired-case interval, and the grouped COVID-19 sensitivity is also negative; neither analysis supports an accuracy-benefit claim for rewriting. Fourth, the current framework audits reliability at dataset and checkpoint level but does not contain a learned per-case confidence gate or automatic image-only fallback. Fifth, continuous translation and zoom in the reported MEDEQUISEG results do not recompute coarse location labels. The completed three-seed MEDEQUISEG + Location Recompute diagnostic decreases BUSI performance and is neutral on ClinicDB, but it is mask derived and does not share exact realized transform draws with the main configuration; it cannot establish the effect of a clinically supplied continuous-language update. Sixth, the public prompts contain attributes extracted from the target masks or labels. This controlled referring benchmark is useful for testing equivariance, but it gives image+text methods privileged target metadata and cannot establish clinical utility or a fair deployment advantage over image-only models. The post hoc stratified derangement controls only

coarse presence and diagnostic class derived from released identifiers/prompts; it does not match morphology, location, wording template, or other target-derived attributes and therefore cannot isolate clinical language grounding. None of the prompts were authored prospectively by clinicians. Seventh, the component study is complete but is an ordered ablation rather than a full crossing of every augmentation-policy interaction. Eighth, although six image-only architectures are evaluated with complete three-seed coverage, architectural recipes are not forced to share input size, loss, or augmentation; the comparison is controlled at the data and evaluation levels rather than as an identical-training ablation. The broad model table is therefore contextual, with rankings separated by modality. Ninth, the five public datasets differ in modality, sample count, prompt diversity, and spatial-language prevalence, so an aggregate average would obscure meaningful heterogeneity. Tenth, the public benchmark manifests do not generally expose patient or acquisition-sequence identifiers. The fixed test folders are image-disjoint, and the recoverable COVID-19 subject IDs do not cross into test, but patient-level independence cannot be established for every dataset from the released metadata. For COVID-19, the grouped train-validation sensitivity removes all overlap among 2,130 recoverable subject tokens and changes MEDEQUISEG Dice by only -0.19 pp, but 2,951 rows without recoverable tokens remain case-level groups. The reported-checkpoint subject-cluster sensitivity preserves the direction of the large U-Net++ and shuffled-prompt contrasts, while the small rewrite contrast remains uncertain; comparable clustered inference is not possible for the other public datasets from the released metadata. ClinicDB sequence independence also cannot be verified because sequence identifiers are absent.

The current manuscript therefore limits its primary claim to a fixed-data-and-evaluation benchmark association led by the shared training policy. Prospective clinician-authored text and independent cross-domain testing remain future validation rather than prerequisites for the reported benchmark comparison. Complete boundary-metric tables, model complexity, and qualitative failure panels are provided in the supplement and main manuscript.

5 Conclusions

This study provides a 90-cell, three-seed audit of shared augmentation and privileged prompts in multimodal medical image segmentation. The complete configuration achieves Public-5 macro Dice of 87.87% and IoU of 80.97% and differs from the horizontal-flip Base reference by $+1.46$ Dice pp (95% CI $+0.95$ to $+1.99$) and $+1.74$ IoU pp (95% CI $+1.13$ to $+2.34$). The shared-plan no-rewrite row is numerically best, and the complete configuration differs from the post hoc Base-plus-plan control by only $+0.04$ Dice pp (95% CI -0.40 to $+0.41$). The supported accuracy claim is therefore attached primarily to the shared follow-up training policy, not to rewriting or to an ATConv4 mechanism. Case-specific target-derived text affects aggregate predictions after coarse presence/class matching, but the effect is markedly attenuated and does not establish clinical-report understanding. The protocol boundary, effective two-projector ATConv graph, negative rewrite evidence, and privileged-prompt control are part of the result, not implementation details to be hidden from interpretation.

Declarations

Ethics approval and consent to participate

This study is a secondary computational analysis of five publicly released, de-identified benchmark datasets. No participants were recruited and no new human data were collected for this study. Ethics approval and consent to participate were therefore not applicable. The analysis used the datasets under the access conditions and ethical dispositions reported by their original providers.

Consent for publication

Not applicable. The manuscript contains no identifiable participant information.

Availability of data and materials

The five public datasets are available from their cited sources. The audited Public-5 Protocol V3 source-and-results package is permanently available at <https://github.com/zjj30/MedEquiSeg>. It contains the protocol lock and registry, sanitized public split manifests with their case-level text prompts, MEDEQUISEG/no-rewrite and baseline code, three-seed result tables and statistical outputs, boundary and complexity source data, qualitative-audit source data, figures, and the manuscript source. Medical images, masks, third-party checkpoints, private-cohort artifacts, and raw training logs are not redistributed.

Competing interests

[TO BE CONFIRMED BY ALL AUTHORS: the authors declare that they have no competing interests.]

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Authors' contributions

[TO BE COMPLETED using author initials for conceptualization, methodology, software, validation, data curation, statistical analysis, writing, supervision, and approval of the final manuscript.]

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Use of generative artificial intelligence

OpenAI Codex was used for code-assisted reproducibility checks and English language editing during manuscript preparation. The authors reviewed and take responsibility for the experimental code, numerical results, citations, and final text. No generative model was treated as an author.

References

- [1] Radford A, Kim JW, Hallacy C, Ramesh A, Goh G, Agarwal S, et al. Learning transferable visual models from natural language supervision. In: Proceedings of the 38th International Conference on Machine Learning. Proc Mach Learn Res. 2021;139:8748–8763.
- [2] Lüddecke T, Ecker AS. Image segmentation using text and image prompts. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2022. p. 7086–7096. doi:10.1109/CVPR52688.2022.00695.
- [3] Zhang S, Xu Y, Usuyama N, Xu H, Bagga J, Tinn R, et al. Biomed-CLIP: a multimodal biomedical foundation model pretrained from fifteen million scientific image–text pairs. arXiv [Preprint]. 2023. arXiv:2303.00915. doi:10.48550/arXiv.2303.00915.
- [4] Koleilat T, Asgariandehkordi H, Nejatimanzari O, Barile B, Xiao Y, Rivaz H. MedCLIPSeg: probabilistic vision–language adaptation for data-efficient and generalizable medical image segmentation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2026. p. 1406–1417.
- [5] Kim W, Son B, Kim I. ViLT: vision-and-language transformer without convolution or region supervision. In: Proceedings of the 38th International Conference on Machine Learning. Proc Mach Learn Res. 2021;139:5583–5594.
- [6] Li Z, Li Y, Li Q, Wang P, Guo D, Lu L, et al. LViT: language meets vision transformer in medical image segmentation. IEEE Trans Med Imaging. 2024;43(1):96–107. doi:10.1109/TMI.2023.3291719.
- [7] Chen Y, Wei M, Zheng Z, Hu J, Shi Y, Xiong S, et al. CausalCLIPSeg: unlocking CLIP’s potential in referring medical image segmentation with causal intervention. In: Medical Image Computing and Computer Assisted Intervention – MICCAI 2024. Lect Notes Comput Sci. 2024;15003:77–87. doi:10.1007/978-3-031-72384-1_8.
- [8] Ronneberger O, Fischer P, Brox T. U-Net: convolutional networks for biomedical image segmentation. In: Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015. Lect Notes Comput Sci. 2015;9351:234–241. doi:10.1007/978-3-319-24574-4_28.

- [9] Zhou Z, Siddiquee MMR, Tajbakhsh N, Liang J. UNet++: redesigning skip connections to exploit multiscale features in image segmentation. *IEEE Trans Med Imaging*. 2020;39(6):1856–1867. doi:10.1109/TMI.2019.2959609.
- [10] Liu Y, Zhu H, Liu M, Yu H, Chen Z, Gao J. Rolling-Unet: revitalizing MLP’s ability to efficiently extract long-distance dependencies for medical image segmentation. *Proc AAAI Conf Artif Intell*. 2024;38(4):3819–3827. doi:10.1609/aaai.v38i4.28173.
- [11] Isensee F, Jaeger PF, Kohl SAA, Petersen J, Maier-Hein KH. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nat Methods*. 2021;18(2):203–211. doi:10.1038/s41592-020-01008-z.
- [12] Ma J, Li F, Wang B. U-Mamba: enhancing long-range dependency for biomedical image segmentation. *arXiv [Preprint]*. 2024. arXiv:2401.04722. doi:10.48550/arXiv.2401.04722.
- [13] Li C, Liu X, Li W, Wang C, Liu H, Liu Y, et al. U-KAN makes strong backbone for medical image segmentation and generation. *Proc AAAI Conf Artif Intell*. 2025;39(5):4652–4660. doi:10.1609/aaai.v39i5.32491.
- [14] Shorten C, Khoshgoftaar TM. A survey on image data augmentation for deep learning. *J Big Data*. 2019;6:60. doi:10.1186/s40537-019-0197-0.
- [15] Yi J, Uzkent B, Ignat O, Li Z, Garg A, Yu X, et al. Augment the Pairs: semantics-preserving image-caption pair augmentation for grounding-based vision and language models. In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 2024. p. 5520–5530.
- [16] Cai K, Yan C, He M, Qu L, Wang S, Tan T. Region-based text-consistent augmentation for multimodal medical segmentation. In: *Medical Image Computing and Computer Assisted Intervention – MICCAI 2025*. *Lect Notes Comput Sci*. 2025;15962:533–543. doi:10.1007/978-3-032-04947-6_51.
- [17] Li A, Zeng X, Zeng P, Ding S, Wang P, Wang C, et al. Textmatch: using text prompts to improve semi-supervised medical image segmentation. In: *Medical Image Computing and Computer Assisted Intervention – MICCAI 2024*. *Lect Notes Comput Sci*. 2024;15008:700–709. doi:10.1007/978-3-031-72111-3_66.
- [18] Wang Q, Lin X, Yan Z. Towards robust medical image referring segmentation with incomplete textual prompts. In: *Medical Image Computing and Computer Assisted Intervention – MICCAI 2025*. *Lect Notes Comput Sci*. 2025;15966:636–646. doi:10.1007/978-3-032-04981-0_60.
- [19] Zhang W, Luo L, He M, Hai J, Ye J. DescriptorMedSAM: language-image fusion with multi-aspect text guidance for medical image segmentation. *Sci Rep*. 2026;16:3758. doi:10.1038/s41598-025-33843-5.

- [20] Liao L, Liu Y, Geng Q. Spatial-Aware Symmetric Alignment for Text-Guided Medical Image Segmentation. In: 2025 IEEE International Conference on Bioinformatics and Biomedicine (BIBM). 2025. p. 3828–3831. doi:10.1109/BIBM66473.2025.11356065.
- [21] Chen X, Wang Y, Pang G, Hao J, Yue C, Zhou L, et al. Medical referring image segmentation via next-token mask prediction. *IEEE Trans Med Imaging*. 2026;Early Access:1–1. doi:10.1109/TMI.2026.3705770.
- [22] Chai S, Jain RK, Xu R, Mo S, Hou R, Teng S, et al. A text-image fusion method with data augmentation capabilities for referring medical image segmentation. *arXiv [Preprint]*. 2025. arXiv:2510.12482. doi:10.48550/arXiv.2510.12482.
- [23] Li Y, Zhao Z, Liu Y, Zhou L. Semi-MedRef: semi-supervised medical referring image segmentation with cross-modal alignment. *arXiv [Preprint]*. 2026. arXiv:2605.15720. doi:10.48550/arXiv.2605.15720.
- [24] Yu H, Chen H, Jiang Y, Peng W, Sun Z, Kaski S, et al. Attentive convolution: unifying the expressivity of self-attention with convolutional efficiency. *arXiv [Preprint]*. 2025. arXiv:2510.20092. doi:10.48550/arXiv.2510.20092.
- [25] Al-Dhabyani A, Gomaa M, Khaled H, Fahmy A. Dataset of breast ultrasound images. *Data Brief*. 2020;28:104863. doi:10.1016/j.dib.2019.104863.
- [26] Bernal J, Sánchez FJ, Fernández-Esparrach G, Gil D, Rodríguez C, Vilariño F. WM-DOVA maps for accurate polyp highlighting in colonoscopy: validation vs. saliency maps from physicians. *Comput Med Imaging Graph*. 2015;43:99–111. doi:10.1016/j.compmedimag.2015.02.007.
- [27] Gómez-Flores W, Gregorio-Calas MJ, Coelho de Albuquerque Pereira W. BUS-BRA: a breast ultrasound dataset for assessing computer-aided diagnosis systems. *Med Phys*. 2024;51(4):3110–3123. doi:10.1002/mp.16812.
- [28] Fateh A, Rezvani Y, Moayedi S, Rezvani S, Fateh F, Fateh M, et al. BRISC: annotated dataset for brain tumor segmentation and classification. *Sci Data*. 2026;13:361. doi:10.1038/s41597-026-06753-y.
- [29] Degerli A, Ahishali M, Yamac M, Kiranyaz S, Chowdhury MEH, Hameed K, et al. COVID-19 infection map generation and detection from chest X-ray images. *Health Inf Sci Syst*. 2021;9:15. doi:10.1007/s13755-021-00146-8.
- [30] Holm S. A simple sequentially rejective multiple test procedure. *Scand J Stat*. 1979;6(2):65–70.